1 Purpose

The prototype TB ABM, was developed to include the basic disease and population mechanisms, agent heterogeneity and architecture, which could form the basis of more complex ABM, with further development.

2 Entities, state variables, and scales

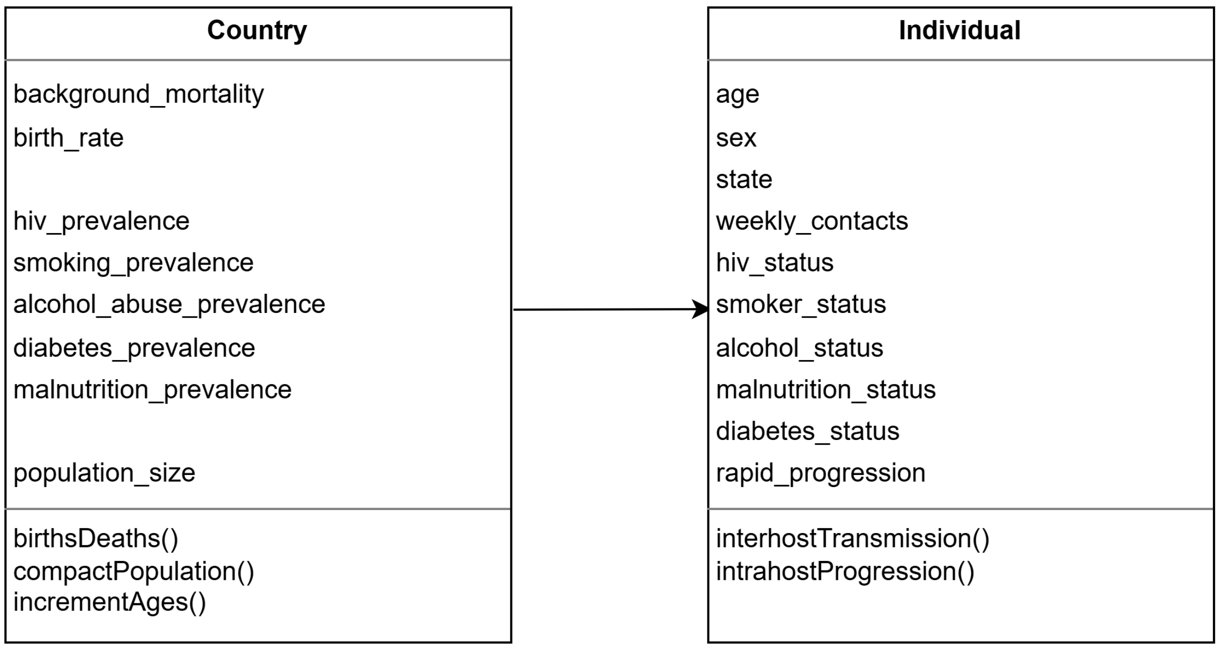

Figure 1: Model entities with attributes and methods

The lowest-level entities in the model are Individuals, which are assigned an age, number of weekly contacts, and a list of comorbidities. Individual agents operate on a weekly time scale and facilitate TB transmission in the model. Agent risk factors are used to calculate the agent's risk of rapid progression to active disease, represented by the rapid progression attribute. Risk factors are assigned to agents during initialisation and remain fixed for the agent's lifetime. Movement through health states is governed by the interhostTransmission and intrahostProgression submodels. The state attribute represents an individual's current position within health states.

The Country agent is a collective agent that represents all individuals within the national population, holds attributes that determine the prevalence of risk factors and manages events that take place on an annual time scale. The Country agent's birth rate and background mortality attributes drive population growth and decline within the model, while the population size attribute tracks the current population size throughout the simulation.

3 Process overview and scheduling

Algorithm 1: Prototype TB ABM simulation schedule

```
Initialise:
    loadData();
    initialiseResults();
    initialisePopulation();
for t ← 0 to numruns do
    for t ← starting_year to final_year do
        for w ← 0 to 51 do
            interHostTransmission();
            intraHostProgression();
        incrementAges();
        birthsDeaths();
        updateRiskFactors();
    reinitialise();
```

As shown in Algorithm 1, model initialisation is completed in three steps. The `loadData()` function extracts crude annual birth and death rates, and total annual population size for the period of 1985-2023 from a CSV. Validation data for annual active and latent TB prevalence and TB mortality for the same period are also loaded in this step. The `initialisePopulation()` function sets the initial and maximum population size, creates all per-agent arrays and initialises agent health states. The `initialiseResults()` function creates all per-year arrays used to capture annual model outputs and validation values.

To facilitate multiple uninterrupted runs, model execution is handled by three loops, representing the number of runs, modelled years and the number of weeks within each year. Time is modelled in discrete weekly and annual steps. This allows processes that occur infrequently or need to be applied to all agents to be grouped and parallelised. Annual steps manage the population through: annual increments to agent ages, the addition of newborn agents to the system, the pruning of deceased agents from state arrays, and the calculation of model outputs. The `interHostTransmission()` and `intraHostProgression()` submodels run on weekly scales, to allow for realistic contact and exposure patterns between agents. The model is reinitialised after each run by zeroing all arrays to avoid repeated memory allocation, and data transfers between the CPU and GPU.

4 Design concepts

4.1 Emergence

There is no true emergent behaviour in the model in its current state. The trajectory of the TB epidemic is the output pattern created by the model, and while no rules are explicitly set about the behaviour of the epidemic, the way that agents move through health states is set programmatically. Agents do not currently have a choice (i.e., start treatment, discontinue treatment, opt to isolate) which would lead to variable agent behaviour.

4.2 Adaptation, objectives, learning, prediction and sensing

Agents do not currently have any self-directed purpose, ability to sense their environment or change their behaviour based on experience.

4.3 Interaction

Agents interact through random contact in the model. Each agent has a set number of 20 weekly contacts. This contact is estimated computationally. Interactions between susceptible and infected individuals probabilistically lead to the spread of TB infection.

4.4 Stochasticity

All agent initialisation and state transitions rely on stochastic random number draws. The model also relies on stochastic Bernoulli trials using probabilities converted to annual hazards to model agent health state changes.

4.5 Collectives

All agents in the model form a collective Country agent, which has a total population size and an annual birth and mortality rate.

4.6 Observation

Data collected from the model include, annual health state totals, cumulative TB deaths per year, active and latent TB prevalence, and annual model runtime.

5 Initialisation

The model is intialised with the South African population size in 1985, which is equal to 35 042 093, with an expected initial latent TB prevalence of 82.70% (of which 77.69% are distant and 5% are recent latent infections) and an active TB prevalence of 0.80%. Health states are initialised based on a random number draw for each agent, which is used to assign each agent to one of four initial health states (0: Susceptible, 1: Recent Latent, 2: Distant Latent). Random number draws are stochastic between runs, but owing to the large population size and the law of large numbers, these end up being roughly the same between runs. Agent risk factors are randomly assigned based on an assumed prevalence of 10%. Risk factors are not mutually exclusive, unlike health states.

6 Input data

Input data include time series data on the annual South African crude birth and mortality rates, sourced from the World Bank Development Indicators Dataset[1]. No other time-varying processes in the model are based on external data.

7 Submodels

The model state is driven by several submodels, which facilitate TB transmission, individual agent progression through health states, as well as background population growth and decline. Similar to the majority of epidemiological ABMs described in Section **??**, the developed ABM makes use of a state-based model for transmission and progression.

Agents can be in one of four health states depicted in Figure 2. The model structure was chosen based on findings in the literature that models, which separate recent and distant latency states, most accurately reproduced TB estimates in validation data. Agents progress through disease states linearly, with the possibility of falling back into previous states, indicated by dotted lines.

Agents in the *Susceptible* state are susceptible to primary infection. Agents in *Recent Latent* and *Distant Latent* states are susceptible to reinfection. The probability of infection and reinfection is determined by an agent's social contact within the model, represented by the number of `weekly_contacts` and the total

[1] https://databank.worldbank.org/source/world-development-indicators

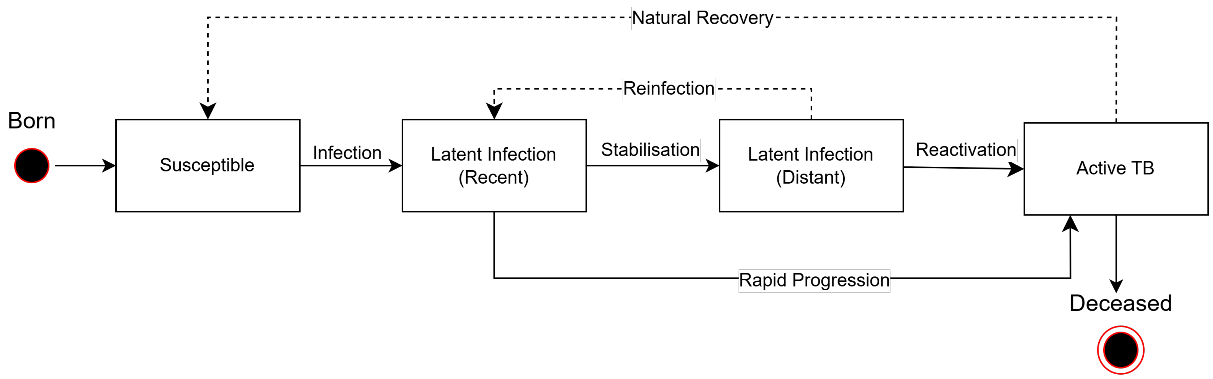

Figure 2: The conceptual model of TB disease progression

actively infected population size. Once latently infected, the probability of rapid progression varies per agent based on the presence of TB risk factors.

Population growth and decline are considered in the model through the addition of newborn agents, as well as TB and non-TB background deaths. Newborn agents are susceptible by default, and are assigned a list of comorbidities at birth.